Step 1: Installation

PIMMS installs from a .exe file available at https://github.com/greg1d/PFAS-IMplementor-for-Mass-Spectrometry--PIMMS-/releases

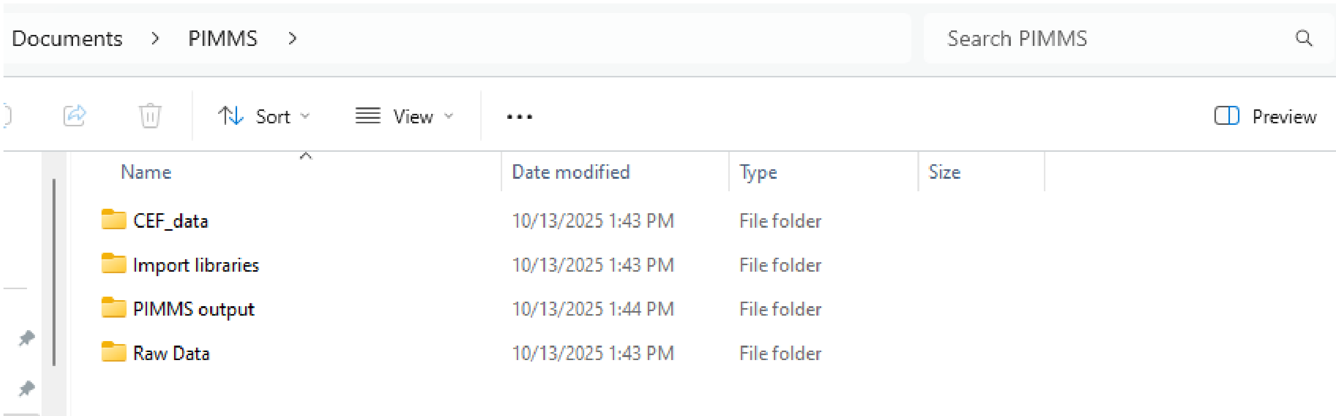

Figure 2 Files created in the user documents folder upon installation. Contained is all data necessary to perform the same method validation study performed within this paper. Isotopic distribution data is contained in “CEF_data”, “Import libraries” contain all data necessary for scoring, “PIMMS output” is a utility folder for the user to save reports to although on installation is empty, “Raw Data” contains the raw feature list used for the method validation study.

On installation, PIMMS will automatically create a folder containing all data that will be needed for running in the Documents folder. Simply go to the Documents folder and look for “PIMMS”. Here, all libraries given in the Supplemental Tables are given. These libraries are included as default options however, the user can modify these libraries or replace them with libraries of their own.

Step 2: Files Path

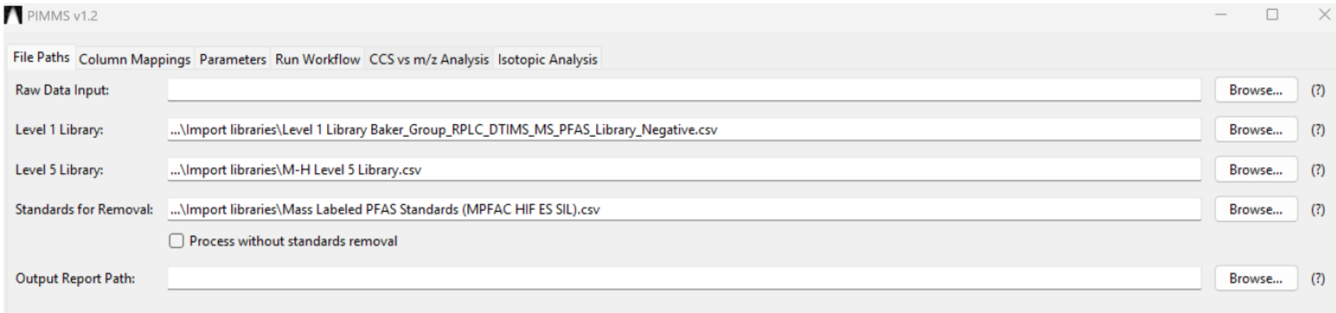

Figure 3 Files Paths tab within PIMMS

Raw Data Input

The raw data input is where the feature list for scoring is imported. The feature list must be a .csv file. This feature list must contain the following columns – ID, *m/z*, CCS, DT, RT, and then the intensities of each sample for each feature. All values outside of the first row (header) must be numeric. The exact name of the header columns does not matter (i.e. CCS could be labeled "Collisional Cross Section"), as the identity of each column is specified in Step 3, Column Mappings. Two sample groups should be present – an experimental and a control population. The control can be any sample type that the user wishes to compare against such as a procedural blank, an unexposed sample population, or a standard blank.

Level 1 Library

The level 1 library corresponds to features which are obtained from standards ran on a comparable LC-IMS-MS platform with a calibrated CCS, *m/z* and RT value. If the RT value was determined using a different chromatography method, this scoring can be deselected in Step 4 Parameters. By default, PIMMS installs and reads in the Baker Lab's PFAS library of 119 features representing 104 unique PFAS for which an analytical standard has been run. The user can update this library locally as more standards come available.

Level 5 Library

The level 5 library corresponds to features that possess a validated structure and *m/z* value; however, no standard is available for further structural confirmation. By default, within PIMMS, a library of 2586 features derived from the EPA Comptox database.[2]

Standards for Removal

Standards for removal corresponds to the features which the user wishes to exclude from the report. By default, PIMMS installs a library of CCS and *m/z* values corresponding to the PFAS analytes within the isotopically labeled mixture MPFAC-HIF-ES.[3] If the user did not add isotopically standards or does not want them to be removed the "Process without standards removal" box can be checked which will ignore any entry in the Standards for Removal box

Output Report Path

This is the location the PIMMS report is outputted to. The name of the report is also specified in this step as well.

Step 3: Column Mappings

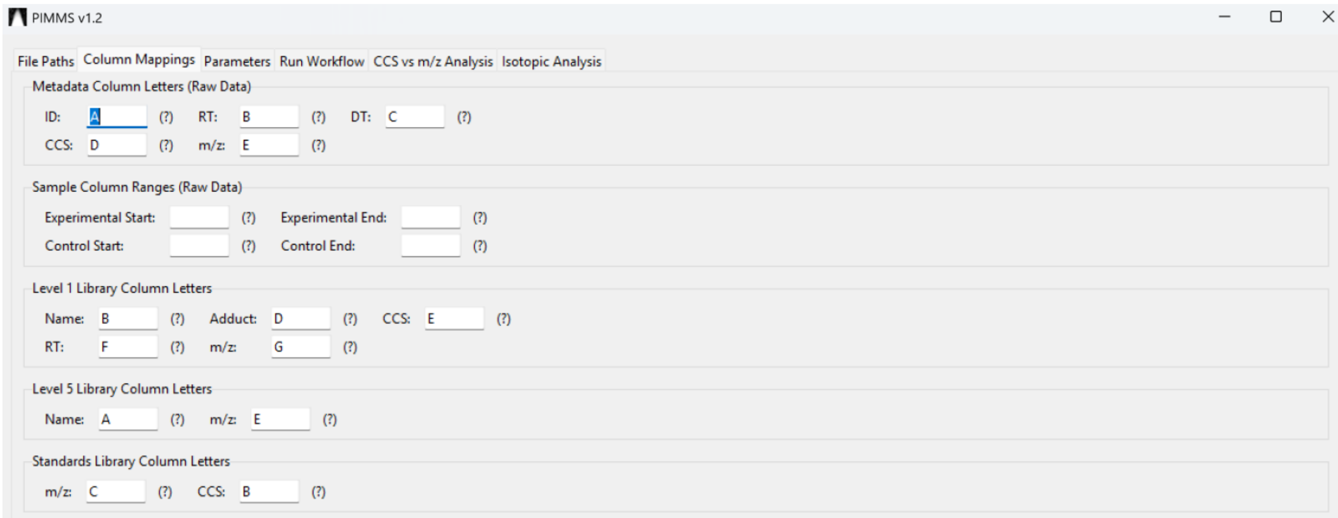

Figure 4 Column Mappings tab within PIMMS containing all the mappings necessary for scoring data.

Metadata Column Letters (Raw Data)

This section allows the user to specify which columns contain what data within the raw feature list. As a default PIMMS is configured to load ID in column A, RT in B, DT in C, CCS in D, and *m/z* in E. By specifying the default columns the user is given the ability to change the name of the column headers as desired.

Sample Column Ranges (Raw Data)

This section allows the user to specify which columns contain the experimental and control population in the feature list. By specifying the columns, the user is given the option to run batches – by separating the columns into sub groups different portions of data can be processed separately without requiring a new feature list. The "start" column corresponds to the first column that the data is contained in, the "end" corresponds to the last.

Level 1 Library Column Letters

This corresponds to the mapping of the Level 1 library. PIMMS by default is configured to work with the Baker Lab's CCS library, however, if another library is used in which the order of the columns is different, simply modify the column mappings to align with the new library.

Level 5 Library Column Letters

This corresponds to the mapping of the suspects library. PIMMS by default is configured to work with the EPA comptox library suspect library, however, if another library is used in which the order of the columns is different, simply modify the column mappings to align with the new library.

Standards Column Library

This corresponds to the mapping of the standards library. PIMMS by default is configured to work with the MPFAC-HIF-ES library, however, if another library is used in which the order of the columns is different, simply modify the column mappings to align with the new library.

Step 4: Parameters

Tolerances and Filters - Mass Error (ppm)

The maximum mass error that a feature can have to be considered a "match" to a standard or suspect. The value is absolute, so 15 ppm would be a range from -15 ppm to + 15 ppm. For a run in which mass errors for known suspects are consistently within -10 - +10 ppm, an expanded error of 15 ppm is suggested as peak alignment algorithms within different feature mining software can adjust *m/z* values by a few ppm.

Tolerances and Filters - CCS Tolerance (%)

The maximum error in the value of CCS that a feature can have to be considered a "match" to a standard. The value is absolute, so a 2% error would be a range from 98% to 102% of the true value.

Tolerances and Filters - RT Tolerance (min)

The maximum error in the value of RT that a feature can have to be considered a "match" to a standard or suspect. The value is absolute, so a tolerance of 0.5 would be a range from -0.5 to +0.5 minutes of the true value.

Tolerances and Filters - Minimum Intensity

The minimum intensity of a feature to be recorded after blank subtraction. Note, intensities in Agilent Mass Profiler are approximately 2% of the peak area in Skyline. Therefore, a peak area threshold of 10,000 in Skyline would correspond to only 200 in PIMMS.

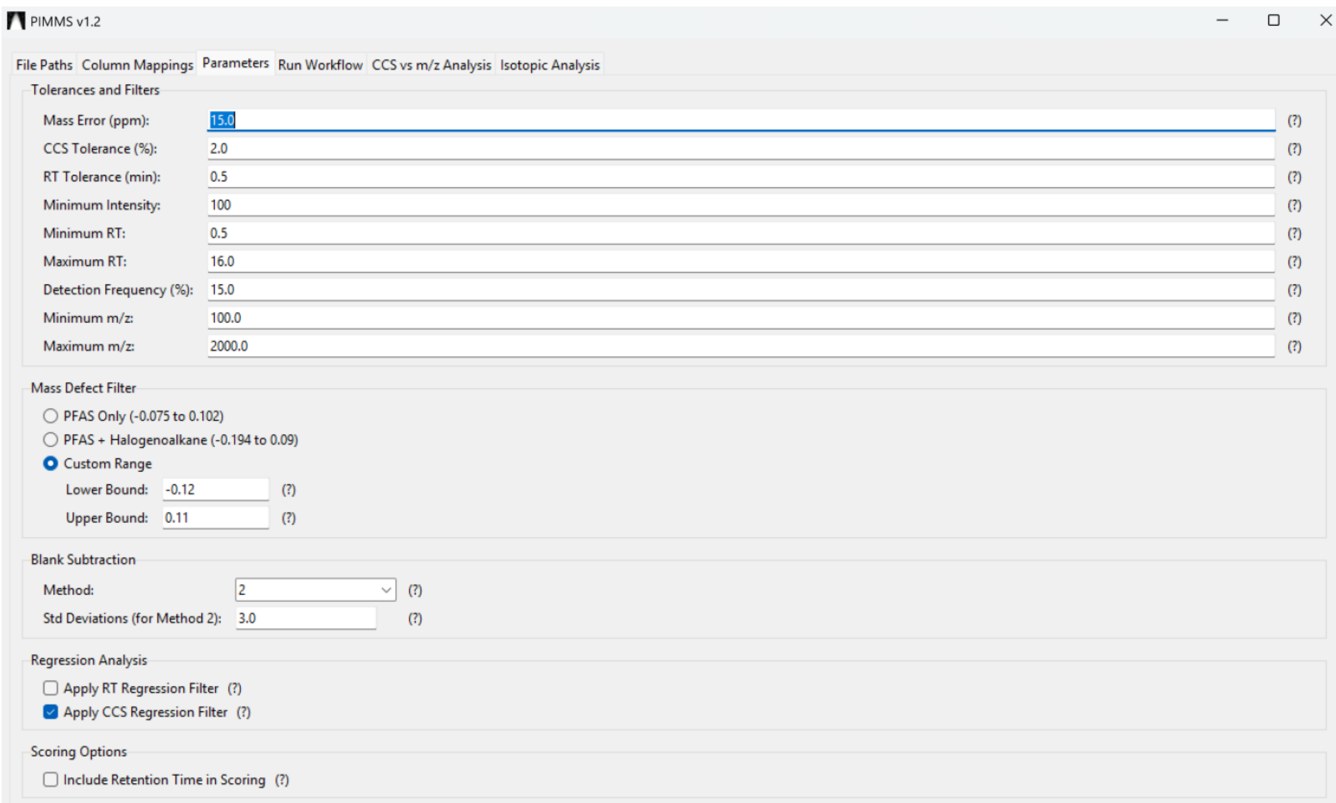

Figure 5 Parameters tab in which all processing parameters are specified to score data.

Tolerances and Filters - Minimum RT, Maximum RT

The retention bounds for a feature in minutes. This allows the user to design batches to their sample processing to consider only certain regions of the chromatography within a single report.

Tolerances and Filters - Detection Frequency (%)

The minimum frequency of a feature must be detected after all filtering steps to be reported. A detection frequency of 20 would correspond to a feature being detected in at least 1/5th of experimental samples after all filtering steps.

Tolerances and Filters - Minimum *m/z*, Maximum *m/z*

The *m/z* bounds for a feature in Da to be reported.

Mass Defect Filter – PFAS only bounds

By selecting this option the user filters features to only those that are within -0.075 – 0.102 Da of the closest integer mass corresponding to the 5^{th} – 95^{th} percentile bounds of the NORMAN PFAS suspect library filtered to only those features detected via ESI – mode and reversed phase chromatography retention according to the NORMAN library's prediction methods.

Mass Defect Filter – PFAS + Halogenoalkane bounds

By selecting this option the user filters features to only those that are within -0.194 – 0.090 Da of the closest integer mass. This value is derived from the 5^{th} – 95^{th} percentile bounds of the NORMAN library taking the same PFAS library as above and combining this with a library of halogenoalkanes containing chlorine, bromine, and iodine.

Mass Defect Filter – Custom bounds

By selecting this option the user filters features to only those that are within the range the user desires. This gives the user the option to customize their workflow to only target features within a specific defect range for enhanced customizability. By default, these values are set to -0.12 to 0.11.

Blank Subtraction – Method 1

By selecting this option, the user subtracts the highest intensity value contained within the control population for each feature from the experimental population. I.e. a control population for a feature of 14, 9, 100, 18, 17 would subtract 100 from every feature in the experimental population for that feature.

Blank Subtraction – Method 2

By selecting this option, the user subtracts the mean + x multiples of standard deviation of the control population from each sample in the experimental population. The box "Std deviations (for Method 2)" allows the user to specify the number of standard deviations that are used in this thresholding. PIMMS defaults to 3 standard deviations to align with standard operating procedures for a limit of detection.[4] However, this value can be set to any value for a more or less conservative threshold.

Regression Analysis – Apply RT Regression Filter

By selecting this feature, the standard library the user selects is used to create a 10^{th} – 90^{th} percentile logarithmic equation to determine where a typical PFAS sits in the RT v *m/z* relationship. The regression filter for RT is limited in application, however. The data must be collected using the same chromatographic conditions as the standards library. Moreover, selecting this filter may increase the false negative rate (reporting PFAS which are present in a sample as not being present) because of sample-specific phenomena which can influence the reported RT such as matrix effects. Therefore, by default, this filter is unchecked, although the user may select it to include it in their NTA workflow.

Regression Analysis – Apply CCS Regression Filter

By selecting this feature, the standard library the user selects is used to create a 10^{th} – 90^{th} percentile logarithmic equation to determine where a typical PFAS sits in the CCS v *m/z* relationship. By default, this filter is selected although may be deselected if desired.

Scoring Options – Include Retention Time in Scoring

By selecting this feature, features matching the standards library are now scored based on their RT in addition to their CCS and *m/z* value. The tolerance for this matching is given in the RT tolerance (min) field.

Step 5: Run Workflow

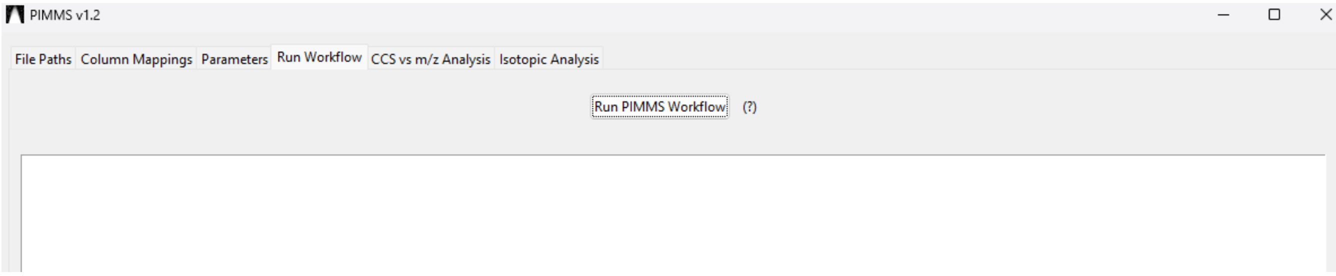

Figure 6 Run workflow tab in which the PIMMS report is created with a terminal showing a read out of the process through the creation of the report as well as any errors in processing.

Run PIMMS Workflow

The user hits the "Run PIMMS Workflow" button to process data. The terminal below the workflow button displays the progress of the report processing.

Step 6: CCS v *m/z* Analysis

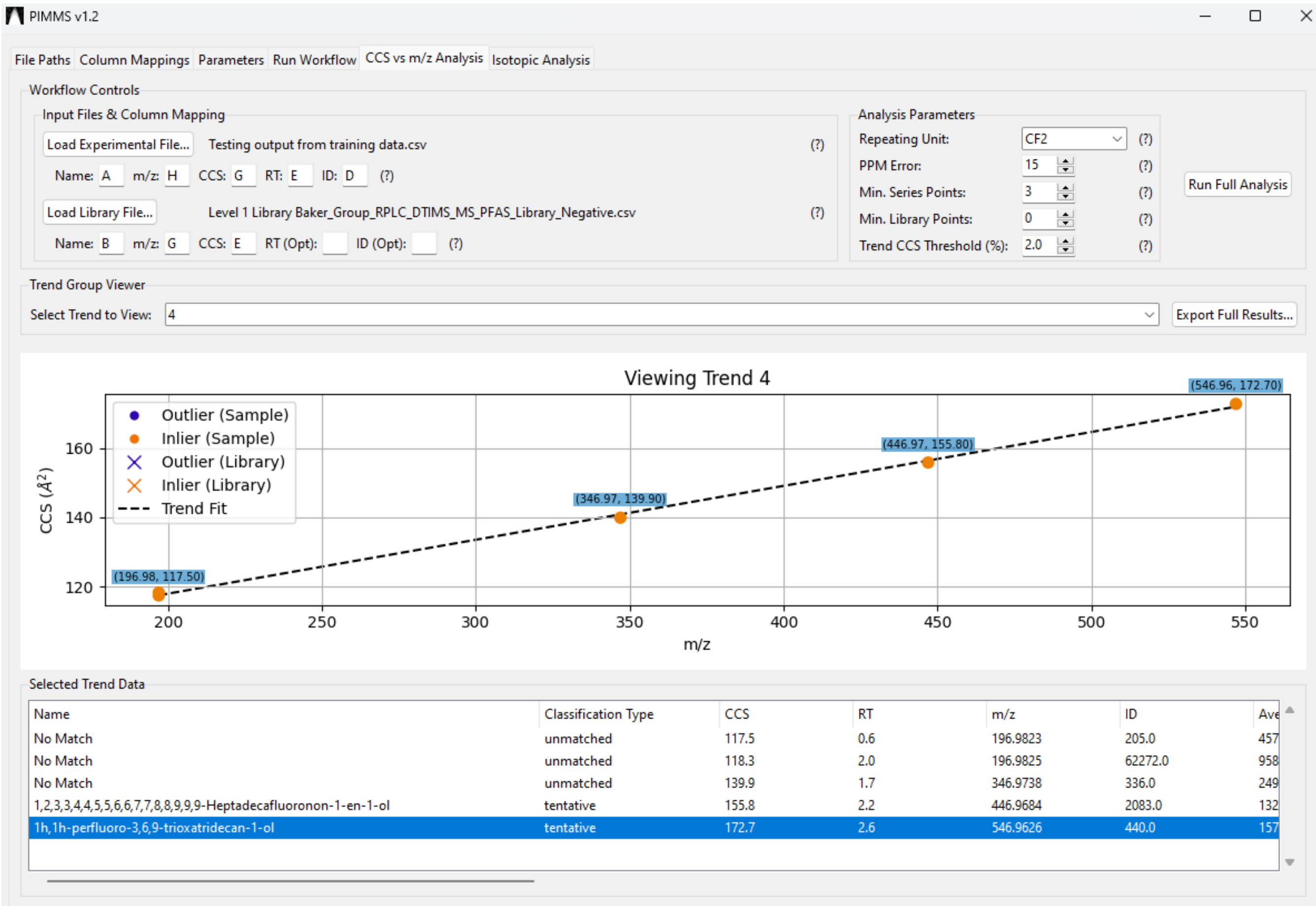

Figure 7 CCS v *m/z* trend line for a series of points matched to PFAS alcohols with 2 compounds being unmatched suggesting them to be novel. As seen in the selected trend data, an overview of all chemical information needed for further investigation is given.

Input Files & Column Mapping – Load Experimental File

The load experimental file button opens a file explorer that gives the user the ability to import any PIMMS report file desired. In addition, this functionality has the added advantage that it is software agnostic – a report from another NTA software or manual analysis can be used so long as the columns "Name", "m/z", "CCS", "RT" and "ID" exist.

Column mappings are set by default to the format that PIMMS exports as, however, if the report has been modified to place the data in a different format, then adjusting the column mappings will maintain the original performance.

Input Files & Column Mapping – Load Library File

The load library file button opens a file explorer that gives the user the ability to import any library file desired. By default, this is set to the Baker Lab PFAS library contained in the user's documents folder. However, the us

Column mappings are set by default to the format that PIMMS exports as, however, if the report has been modified to place the data in a different format, then adjusting the column mappings will maintain the original performance.

Analysis Parameters – Repeat Unit

The homologous series which the user searches against – 5 options are made available: CF_2, OCF_2, OC_2F_4, OC_3F_6, C_2F_3Cl.

Analysis Parameters – PPM error

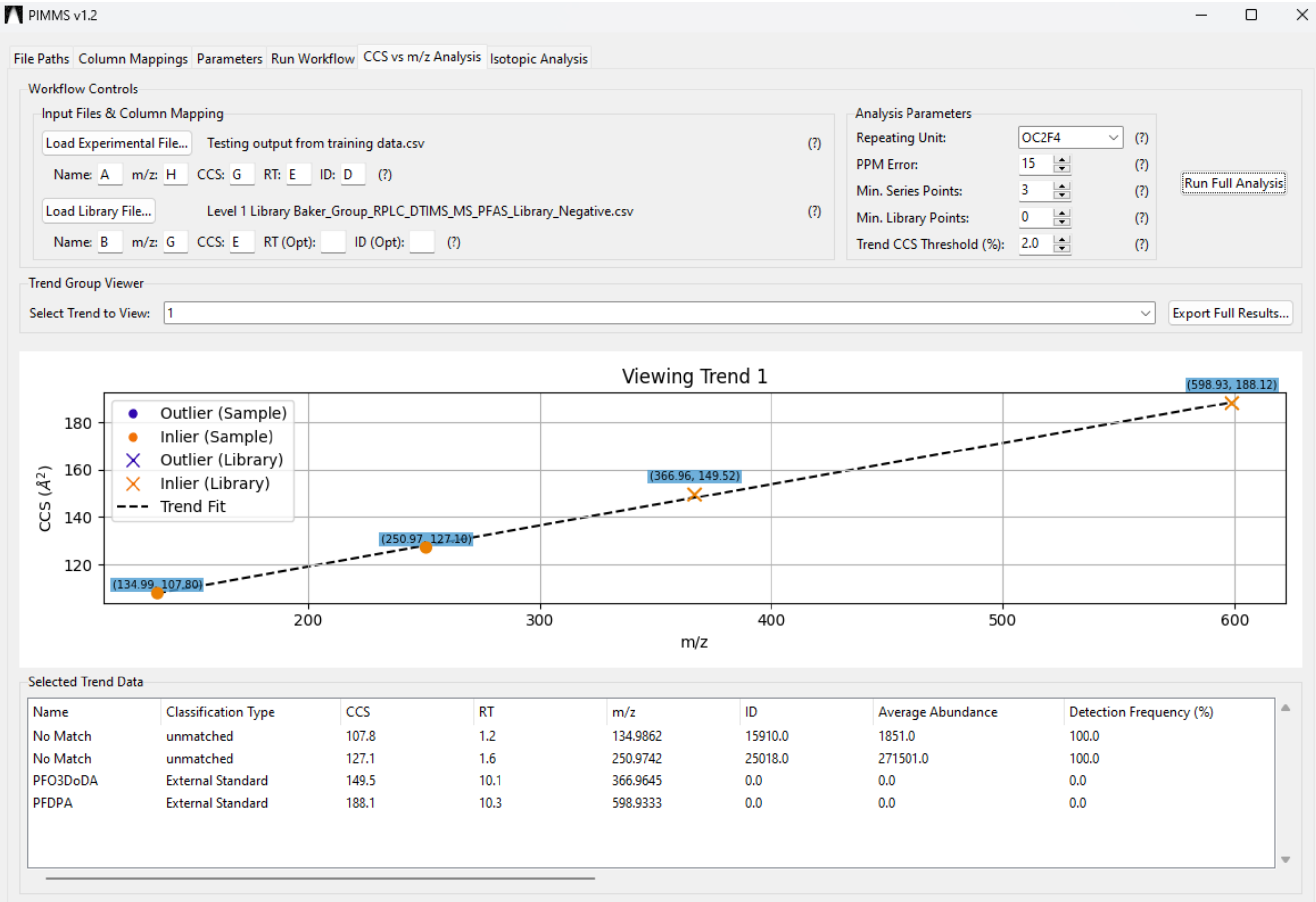

Figure 8 CCS v *m/z* trend line for a series of points matched to standards within the library using the OC_2F_4 repeating unit. Circles denote points originating from the PIMMS report (samples) and crosses denote points coming from the uploaded library. As can be seen 2 points lying in series with PFO3DoDA and PFDPA are unmatched, suggesting them to be potentially novel PFAS that would be otherwise unidentified given their low abundances.

The maximum mass error that a feature can have to be considered a "match" to a theoretical value for a homologous series search. The value is absolute, so 15 ppm would be a range from -15 ppm to + 15 ppm.

Analysis Parameters – Min. Series Points

The minimum number of points that must be present in a series to be reported. PIMMS has a minimum of 3, however, the user can select any value higher than this.

Analysis Parameters – Min. Library Points

The minimum number of points within a given series that must originate from the library file that is uploaded in the load library file step. PIMMS defaults to 0 however any positive value can be used.

Analysis Parameters – Trend CCS Threshold (%)

The maximum error in CCS a point can have from the line of best fit in order to be considered a trend inlier. If the point is in a homologous series trend based on the repeating unit but not in CCS, the point will be reported in blue denoting an outlier. PIMMS defaults to 2% for this value, however, any value can be specified depending on the application.

Analysis Parameters – Export Full Results

The user gets the option to export a report as a .csv of any homologous series that are present in the data. When selected a dialog box opens in which the user can specify any location and name for the file. Within the report is the “trend group” which is an ID number corresponding to each unique homologous series. Also reported is the homologous series repeating unit (i.e. CF_2).

<u>Inlier v outlier in CCS vs *m/z* analysis</u>

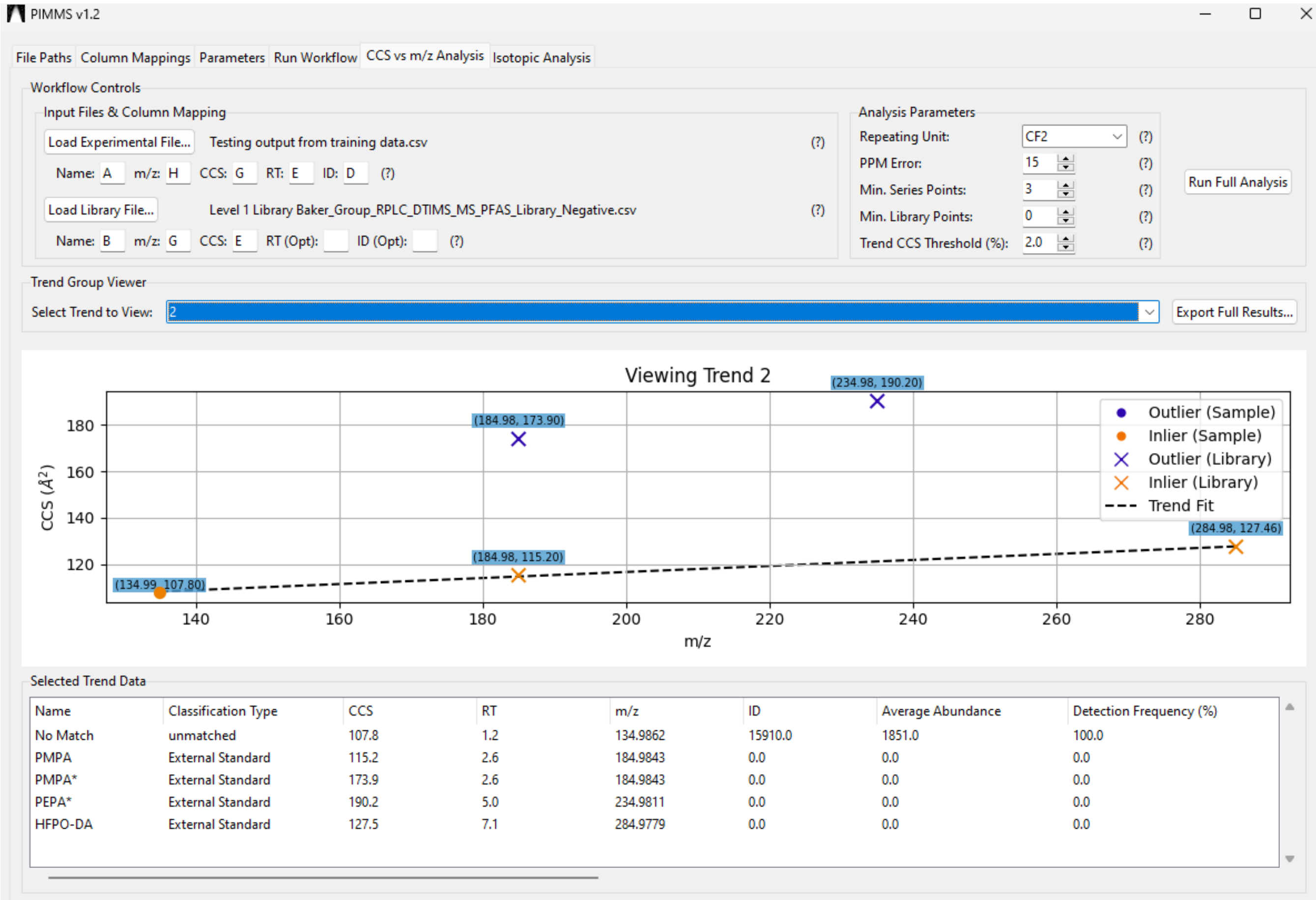

Figure 9 CCS vs. *m/z* trend line for a homologous series containing PMPA and HFPO-DA. Yellow points indicate features within the CCS trend threshold, while blue points denote outliers. Although the PMPA and PEPA library standards fall within the mass-based repeat unit group, they exhibit substantially higher CCS values than predicted by the trend. This deviation correctly identifies these species as dimers, distinct from the monomeric trend defined by HFPO-DA and PMPA.

Step 7: Isotopic Analysis

Workflow – PIMMS File

The browse button opens a file explorer that gives the user the ability to import any PIMMS report file desired.

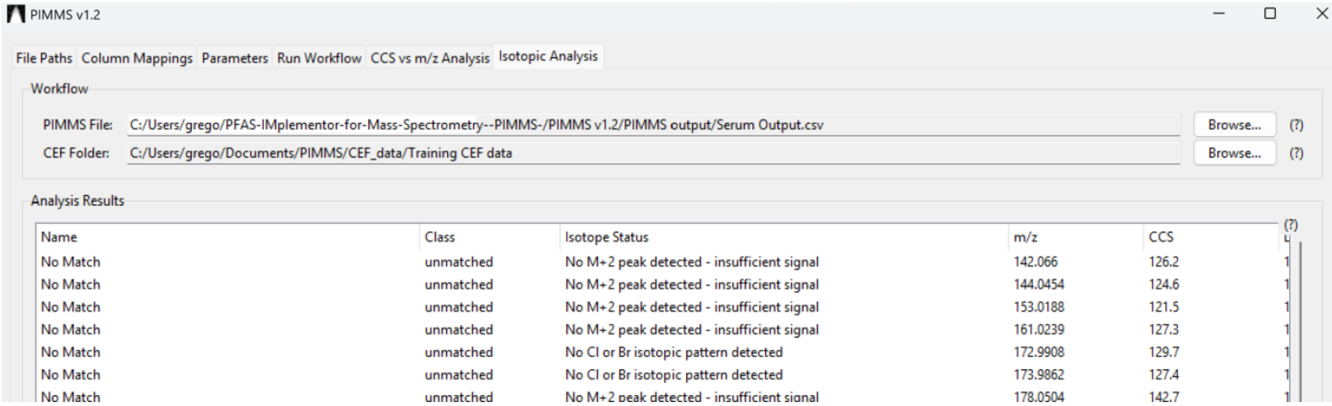

Figure 10 Isotopic analysis tab displaying the import stage and the Analysis Results window.

Workflow – CEF File

The browse button opens a file explorer that gives the user the ability to read in a folder containing the compound exchange files (CEF). The folder can contain different file types and does not need to be only the data within the PIMMS file; PIMMS will identify what files overlap with the samples processed in the PIMMS report and perform the analysis on them.

Run Full Pipeline & Analysis

Performs isotopic analysis on the PIMMS file uploaded and the CEF folder selected.

Analysis Results – Isotope Status

The isotope status field gives an analysis of the M/M+2 peak ratio. If the M+2 peak is at least 28% of the M peak, the result "potential Cl, Br present" is given. If the M+2 peak is <28% of the M peak the result "No Cl or Br isotopic pattern present". If no M+2 peak is detected the result "No M+2 peak detected – insufficient signal" is given.

Analysis Results – Pred. F/C Ratio

The pred. F/C ratio gives an estimate of the ratio of fluorine to carbons within the element using the isotopic distribution of the M+1 peak to the M peak.[5]

Generate Kaufmann Plot

Creates an interactive Kaufmann plot within the user’s browser.

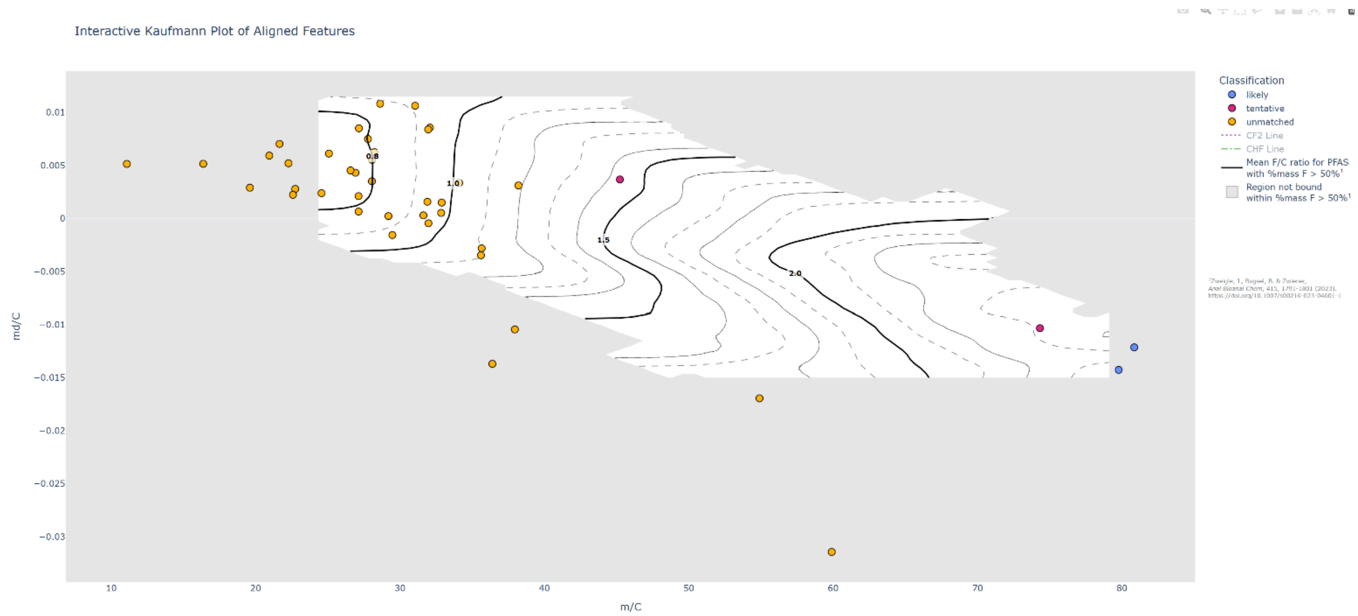

Figure 11 Kaufmann plot of features identified by the isotopic analysis workflow. The Kaufmann plot displays the mass (m) of the M peak divided by the estimated number of carbons “C” taking the ratio of the M:M+1 peak multiplied by 111.45 on the x axis, and the mass defect (md) of the M peak divided by C. From the position on this graph the ratio of fluorines to carbons within each feature can be estimated with the estimate given in the topographical lines.

Within the plot, points are interactive and can be hovered over to reveal key chemical identifier information as well as sample abundances. Points are color coded based on their match to different libraries – the “likely” points are blue matching to the standards library, the “tentative” pink points are points that match only to the user’s Level 5 suspects library. The unmatched yellow points are points that possess no library match.

The contour lines correspond to the predicted F/C ratio with a greater degree of fluorination down and right in the plot at m/C increases and md/C becomes more negative. The contour lines are adapted from Zweigle *et. al*. with the grey region of the plot being area in which the Zweigel data did not cover suggesting this region to have a unique isotopic distribution relative to the data the model was built on such as containing Cl or Br, or not containing a sufficient degree of fluorination.

References

(1) Dodds, J. N.; Hopkins, Z. R.; Knappe, D. R. U.; Baker, E. S. Rapid Characterization of Per- and Polyfluoroalkyl Substances (PFAS) by Ion Mobility Spectrometry–Mass Spectrometry (IMS-MS). *Anal. Chem.* **2020**, *92* (6), 4427–4435. https://doi.org/10.1021/acs.analchem.9b05364.

(2) *CompTox Chemicals Dashboard*. https://comptox.epa.gov/dashboard/chemical-lists/PFASSTRUCT (accessed 2024-01-18).

(3) DCatalog. *Wellington Laboratories Catalogue 2021-2023*. DCatalog. https://wellington-laboratories.dcatalog.com/?docid=05faf8e8-364f-430e-8fcc-1a587c9b7c18&referer=https://well-labs.com/wellingtoncatalogue2123.html&use_externalpopup=1&staging=false (accessed 2025-06-17).

(4) Taylor, J. K. Quality Assurance of Chemical Measurements. *Anal. Chem.* **1981**, *53* (14), 1588A-1596A.

(5) Zweigle, J.; Bugsel, B.; Zwiener, C. Efficient PFAS Prioritization in Non-Target HRMS Data: Systematic Evaluation of the Novel MD/C-m/C Approach. *Anal. Bioanal. Chem.* **2023**, *415* (10), 1791–1801. https://doi.org/10.1007/s00216-023-04601-1.